



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

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22GE004 - BASICS OF ELECTRONICS ENGINEERING

UNIT III - Signal Switching using BJT

1. Transistor as a switch

A transistor can be used as a solid state switch. If the transistor is operated in the saturation region then it acts as closed switch and when it is operated in the cut off region then it behaves as an open switch.

The transistor operates as a Single Pole Single Throw (SPST) solid state switch. When a zero input signal applied to the base of the transistor, it acts as an open switch. If a positive signal applied at the input terminal, then it acts like a closed switch.

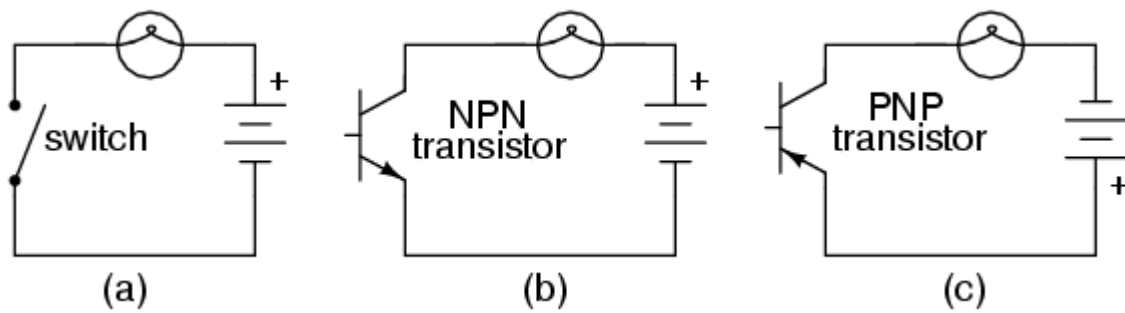
When the transistor operating as switch, in the cut off region the current through the transistor is zero and voltage across it is maximum, and in the saturation region the transistor current is maximum and voltage across is zero. Therefore, both the on – state and off – state power loss is zero in the transistor switch.

Suppose we had a lamp that we wanted to turn on and off with a switch. Such circuit would be extremely simple as in Figure below(a).

For the sake of illustration, let's insert a transistor in place of the switch to show how it can control the flow of electrons through the lamp. Remember that the controlled current through a transistor must go between collector and emitter.

Since it is the current through the lamp that we want to control, we must position the collector and emitter of our transistor where the two contacts of the switch were. We must also make sure that the lamp's current will move *against* the

direction of the emitter arrow symbol to ensure that the transistor's junction bias will be correct as in Figure below(b).

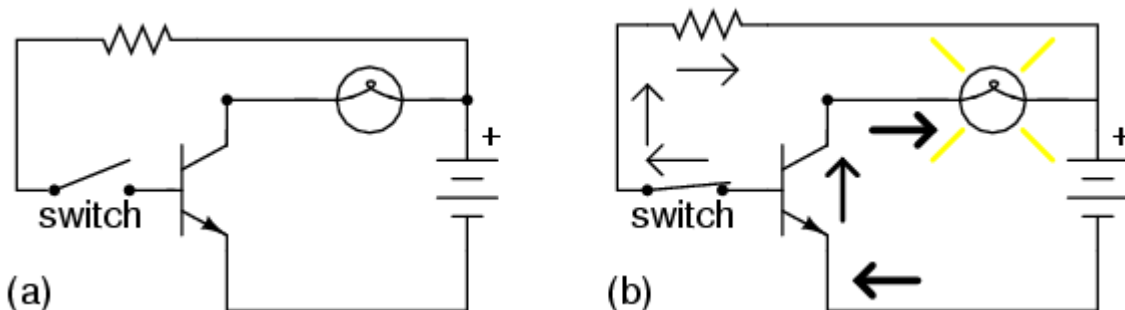


(a) mechanical switch, (b) NPN transistor switch, (c) PNP transistor switch.

A PNP transistor could also have been chosen for the job. Its application is shown in the figure above (c). Note that the battery terminal voltages have been flipped from (b) to (c) to support the replacement of the NPN transistor with a PNP transistor.

The choice between NPN and PNP is really arbitrary. All that matters is that the proper current directions are maintained for the sake of correct junction biasing (current flow going with the transistor symbol's arrow).

In the above figures, the base of either BJT is not connected to a suitable voltage, and no current is flowing through the base. Consequently, the transistor cannot turn on. Perhaps, the simplest thing to do would be to connect a switch between the base and collector wires of the transistor as in figure (a) below.



1.1 Cutoff vs Saturated Transistors

If the switch is open as in (Figure [above](#) (a)), the base wire of the transistor will be left “floating” (not connected to anything) and there will be no current through it. In this state, the transistor is said to be *cutoff*. If the switch is closed as in (Figure [above](#) (b)), however, electrons will be able to flow from the emitter through to the base of the transistor, through the switch and up to the leftside of the lamp, back to the positive side of the battery. This base current will enable a much larger flow of electrons from the emitter through to the collector, thus lighting up the lamp. In this state of maximum circuit current, the transistor is said to be *saturated*.

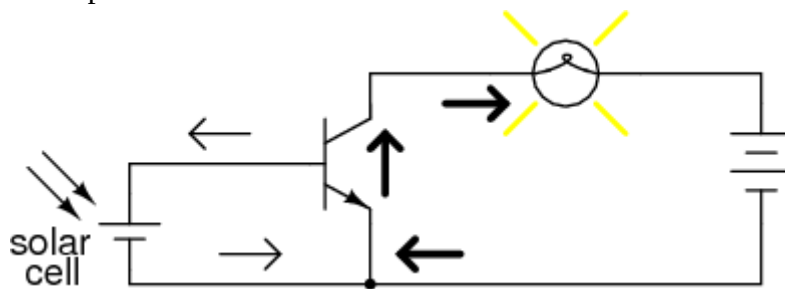
Of course, it may seem pointless to use a transistor in this capacity to control the lamp.

After all, we're still using a switch in the circuit, aren't we? If we're still using switch to control the lamp if only indirectly -- then what's the point of having transistor to control the current? Why not just go back to our original circuit and use the switch directly to control the lamp current?

1.2 Why Use a Transistor to Control Current?

Two points can be made here, actually. First is the fact that when used in this manner, the switch contacts need only handle what little base current is necessary to turn the transistor on; the transistor itself handles most of the lamp's current.

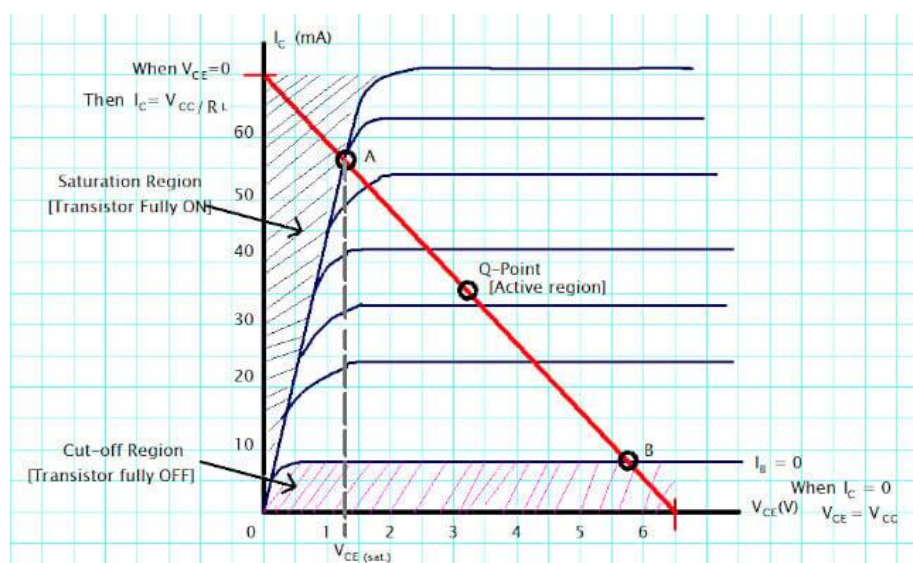
This may be an important advantage if the switch has a low current rating: a small switch maybe used to control a relatively high-current load. More important, the current-controlling behavior of the transistor enables us to use something completely different to turn the lamp on or off. Consider Figure below, where a pair of solar cells provides 1 V to overcome the 0.7 V_{BE} of the transistor to cause base current flow, which in turn controls the lamp.



Solar cell serves as a light sensor.

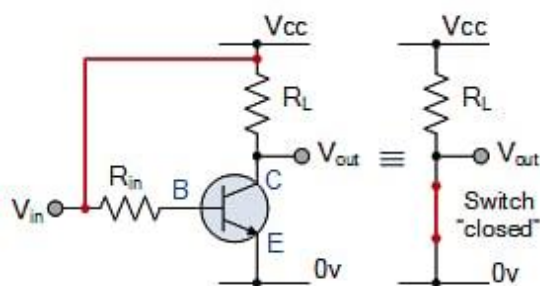
1.3 Transistor Switch Operations and Operating Regions

The blue shaded area on the graph in Figure 1 represents the saturation region, while the pink shaded area represents the cut-off region.



The regions are defined as:

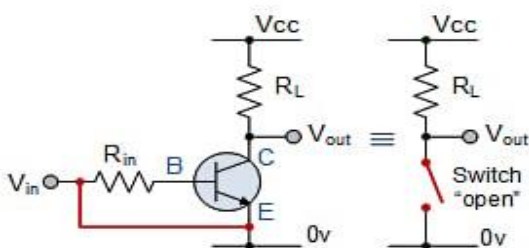
- Saturation region. This is the region where the transistor will be biased for the maximum amount of base current to be used to achieve maximum current at the collector and minimum voltage drop at the collector-emitter and which in turn makes the depletion layer as tiny as possible hence the flow of maximum current through our transistor. This makes the transistor switch ON. Figure 2 demonstrates the saturation region characteristics.



- The input and Base are connected to V_{CC}
- Base-Emitter voltage $V_{BE} > 0.7v$
- Base-Emitter junction is forward biased
- Base-Collector junction is forward biased
- Transistor is "fully-ON" (saturation region)
- Max Collector current flows ($I_C = V_{CC}/R_L$)
- $V_{CE} = 0$ (ideal saturation)
- $V_{OUT} = V_{CE} = "0"$
- Transistor operates as a "closed switch"

Cut-off region. Here the input base current and the output collector current are zero and the collector voltage is at the maximum resulting in a large depletion layer with zero current flowing through the transistor. In such conditions, the transistor is switched off.

The cut-off region characteristics are shown below:



- The input and Base are grounded ($0v$)
- Base-Emitter voltage $V_{BE} < 0.7v$
- Base-Emitter junction is reverse biased
- Base-Collector junction is reverse biased
- Transistor is "fully-OFF" (Cut-off region)
- No Collector current flows ($I_C = 0$)
- $V_{OUT} = V_{CE} = V_{CC} = "1"$
- Transistor operates as an "open switch"

DISCUSSION QUESTIONS:

1. Identify and explain the different regions of operation in a transistor when used as a switch.
2. Illustrate the concept of "load" in a transistor switch circuit.
3. Identify the role of the base current and voltage in controlling the switching action.